



Lab Sheet 2

Database Management System Lab (BTCs203P)

B. TECH AI&ML, CYBER SECURITY, Data Science & Semester(Session 2026-2025)

Objective:

To understand and practice Data Manipulation Language (DML) commands for creating and modifying the structure of a table in a relational database using SQL (MySQL).

Prerequisites:

1. Basic knowledge of SQL.
2. MySQL installed and set up.

Lab Sheet 2: DML Commands– INSERT, UPDATE and DELETE

Perform the following SQL tasks using suitable examples:

1. Insert one record into the Student table.

```
1 CREATE DATABASE COE UNIVERSITY;
2 USE coe_university;
3 CREATE TABLE Student (Student_ID INT PRIMARY KEY, Name VARCHAR(50), Gender VARCHAR(10), Age INT, Address VARCHAR(100),
4 Course VARCHAR(50), Percentage DECIMAL(5,2), PhoneNo VARCHAR(15));
5 SHOW TABLES;
6 DESCRIBE Student;
7 INSERT INTO Student(Student_ID, Name, Gender, Age, Address, Course, Percentage, PhoneNo)
8 VALUES(101, 'Anshika', 'Female', 20, 'Jaipur', 'B.Tech CSE', 85.50, '9876543210');
9 SELECT * FROM Student;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
101	Anshika	Female	20	Jaipur	B.Tech CSE	85.50	9876543210

2. Insert five student records using separate INSERT statements.

```
10 INSERT INTO Student(Student_ID, Name, Gender, Age, Address, Course, Percentage, PhoneNo)
11 VALUES(102, 'Radha', 'Female', 10, 'Kashipur', 'B.Tech CSE', 85.50, '9876543276');
12 INSERT INTO Student(Student_ID, Name, Gender, Age, Address, Course, Percentage, PhoneNo)
13 VALUES(103, 'Jayshree', 'Female', 17, 'Jaipur', 'B.Tech AIML', 85.50, '9876543276');
14 INSERT INTO Student(Student_ID, Name, Gender, Age, Address, Course, Percentage, PhoneNo)
15 VALUES(104, 'Ruhani', 'Female', 20, 'Kashmir', 'B.Tech CSE', 85.50, '9876543276');
16 INSERT INTO Student(Student_ID, Name, Gender, Age, Address, Course, Percentage, PhoneNo)
17 VALUES(105, 'Shreya', 'Female', 20, 'Jaipur', 'B.Tech CSE', 85.50, '9876543276');
18 INSERT INTO Student(Student_ID, Name, Gender, Age, Address, Course, Percentage, PhoneNo)
19 VALUES(106, 'Maya', 'Female', 20, 'Rajasthan', 'B.Tech CSE', 85.50, '9876543276');
20 SELECT * FROM Student;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
101	Anshika	Female	20	Jaipur	B.Tech CSE	85.50	9876543210
102	Radha	Female	10	Kashipur	B.Tech CSE	85.50	9876543276
103	Jayshree	Female	17	Jaipur	B.Tech AIML	85.50	9876543276
104	Ruhani	Female	20	Kashmir	B.Tech CSE	85.50	9876543276
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	Rajasthan	B.Tech CSE	85.50	9876543276

3. Insert multiple records into the Student table using a single statement.

```
23 (108, 'Priya', 'Female', 20, 'Mumbai', 'BCA', 91.00, '9876543212'),
24 (109, 'Aman', 'Male', 22, 'Pune', 'B.Tech IT', 74.75, '9876543213'),
25 (110, 'Neha', 'Female', 21, 'Kolkata', 'B.Tech CSE', 88.50, '9876543214'),
26 (111, 'Rohit', 'Male', 20, 'Chandigarh', 'BCA', 69.25, '9876543215'),
27 (112, 'Simran', 'Female', 22, 'Amritsar', 'B.Tech IT', 82.00, '9876543216');
28 SELECT * FROM Student;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
101	Anshika	Female	20	Jaipur	B.Tech CSE	85.50	9876543210
102	Radha	Female	10	Kashipur	B.Tech CSE	85.50	9876543276
103	Jayshree	Female	17	Jaipur	B.Tech AIML	85.50	9876543276
104	Ruhani	Female	20	Kashmir	B.Tech CSE	85.50	9876543276
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	Rajasthan	B.Tech CSE	85.50	9876543276
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214

4. Insert records into the Employee table.

```
28 • SELECT * FROM Student;
29 • CREATE TABLE Employee(Employee_ID INT PRIMARY KEY,Name VARCHAR(50),Salary DECIMAL(10,2),Department VARCHAR(50),City VARCHAR(50));
30 • INSERT INTO Employee VALUES(11, 'Amit', 35000, 'IT', 'Delhi'),(12, 'Rahul', 42000, 'HR', 'Jaipur'),
31 (13, 'Neha', 50000, 'Finance', 'Mumbai'),(14, 'Priya', 38000, 'IT', 'Pune'),(15, 'Rohit', 28000, 'HR', 'Kota');
32 • SELECT * FROM Employee;
```

Employee_ID	Name	Salary	Department	City
11	Amit	35000.00	IT	Delhi
12	Rahul	42000.00	HR	Jaipur
13	Neha	50000.00	Finance	Mumbai
14	Priya	38000.00	IT	Pune
15	Rohit	28000.00	HR	Kota

5. Insert records into the Department table.

```
32 • SELECT * FROM Employee;
33 • CREATE TABLE Department (Department_ID INT PRIMARY KEY,Department_Name VARCHAR(50),Location VARCHAR(50));
34 • INSERT INTO Department VALUES(101, 'IT', 'Delhi'),(102, 'HR', 'Jaipur'),(103, 'Finance', 'Mumbai'),
35 (104, 'Marketing', 'Pune'),(105, 'Sales', 'Kota');
36 • SELECT * FROM Department;
```

Department_ID	Department_Name	Location
101	IT	Delhi
102	HR	Jaipur
103	Finance	Mumbai
104	Marketing	Pune
105	Sales	Kota

6. Insert records into the Product table.

```
36 • SELECT * FROM Department;
37 • CREATE TABLE Product (Product_ID INT PRIMARY KEY,Product_Name VARCHAR(50),Price DECIMAL(10,2),Quantity INT);
38 • INSERT INTO Product VALUES(1, 'Laptop', 55000, 10),(2, 'Mouse', 800, 25),
39 (3, 'Keyboard', 1500, 15),(4, 'Monitor', 12000, 0),(5, 'Printer', 15000, 5);
40 • SELECT * FROM Product;
```

Product_ID	Product_Name	Price	Quantity
1	Laptop	55000.00	10
2	Mouse	800.00	25
3	Keyboard	1500.00	15
4	Monitor	12000.00	0
5	Printer	15000.00	5

7. Insert records into the Customer table.

```
41 • CREATE TABLE Customer (Customer_ID INT PRIMARY KEY,Name VARCHAR(50),City VARCHAR(50),PhoneNo VARCHAR(15));
42 • INSERT INTO Customer VALUES(1, 'Ankit', 'Delhi', '9876500001'),(2, 'Simran', 'Jaipur', '9876500002'),
43 (3, 'Pooja', 'Mumbai', '9876500003'),(4, 'Aman', 'Pune', '9876500004'),(5, 'Kavya', 'Kota', '9876500005');
44 • SELECT * FROM Customer;
```

Customer_ID	Name	City	PhoneNo
1	Ankit	Delhi	9876500001
2	Simran	Jaipur	9876500002
3	Pooja	Mumbai	9876500003
4	Aman	Pune	9876500004
5	Kavya	Kota	9876500005

8. Insert records into the Course table.

```
45 • CREATE TABLE Course (Course_ID INT PRIMARY KEY,Course_Name VARCHAR(50),Fees DECIMAL(10,2),Duration VARCHAR(30));
46 • INSERT INTO Course VALUES(1, 'B.Tech CSE', 85000, '4 Years'),(2, 'BCA', 60000, '3 Years'),
47 (3, 'B.Tech IT', 80000, '4 Years'),(4, 'MCA', 75000, '2 Years'),(5, 'MBA', 70000, '2 Years');
48 • SELECT * FROM Course;
```

Course_ID	Course_Name	Fees	Duration
1	B.Tech CSE	85000.00	4 Years
2	BCA	60000.00	3 Years
3	B.Tech IT	80000.00	4 Years
4	MCA	75000.00	2 Years
5	MBA	70000.00	2 Years

9. Insert a student record with values for all columns.

```

9 • INSERT INTO Student(Student_ID, Name, Gender, Age, Address, Course, Percentage, PhoneNo)VALUES
0 (114, 'Aditi', 'Female', 21, 'Jaipur', 'B.Tech CSE', 90.50, '9876543250');
1 • SELECT * FROM Student WHERE Student_ID=114;
2

```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250

10. Insert a student record by specifying only selected columns.

```

52 • INSERT INTO Student(Student_ID, Name, Age, Course)VALUES(141, 'Arjun', 21, 'BCA');
53 • SELECT * FROM Student WHERE Student_ID=141;

```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
141	Arjun	NULL	21	NULL	BCA	NULL	NULL

11. Update the city of a particular student.

```

55 • SELECT * FROM Student WHERE Student_ID=101;

```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210

12. Update the salary of a particular employee.

```

56 • UPDATE Employee SET Salary=40000 WHERE Employee_ID = 4;
57 • SELECT * FROM Employee WHERE Employee_ID=4;

```

Employee_ID	Name	Salary	Department	City
4	Priya	45980.00	IT	Pune

13. Increase the salary of all employees by 10%.

```

57 • SELECT * FROM Employee WHERE Employee_ID=4;
58 • UPDATE Employee SET Salary = Salary * 1.10 WHERE Employee_ID > 0;

```

Employee_ID	Name	Salary	Department	City
1	Amit	42350.00	IT	Delhi
2	Rahul	50820.00	HR	Jaipur
3	Neha	60500.00	Finance	Mumbai
4	Priya	45980.00	IT	Pune
5	Rohit	33880.00	HR	Kota

14. Increase the price of all products by 5%.

```

59 • UPDATE Product SET Price = Price * 1.05 WHERE Product_ID > 0;
60 • SELECT * FROM Employee;
61

```

Employee_ID	Name	Salary	Department	City
1	Amit	42350.00	IT	Delhi
2	Rahul	50820.00	HR	Jaipur
3	Neha	60500.00	Finance	Mumbai
4	Priya	45980.00	IT	Pune
5	Rohit	33880.00	HR	Kota

15. Change the course of a particular student.

```

61 • UPDATE Student SET Course = 'B.Tech IT' WHERE Student_ID = 102;
62 • SELECT * FROM Student;

```

Result Grid

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
103	Jayshree	Female	17	Jaipur	B.Tech AIML	85.50	9876543276
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250
141	Arjun	MALE	21	MALE	BCA	MALE	MALE

16. Update the department of an employee.

64 • `SELECT * FROM Employee ;`

65 • `UPDATE Student SET Address = 'Delhi' WHERE Address = 'Meerut';`

result Grid |   Filter Rows: | Edit:    | Export/Import:  

Employee_ID	Name	Salary	Department	City
11	Amit	35000.00	IT	Delhi
12	Rahul	42000.00	HR	Jaipur
13	Neha	50000.00	Finance	Mumbai
14	Priya	38000.00	IT	Pune
15	Rohit	28000.00	HR	Kota

17. Update the city of all students from Meerut to Delhi.

63 • UPDATE Student SET Address = 'Delhi' WHERE Address = 'Meerut';

64 • SELECT * FROM Student;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: ☒

	Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
▶	101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210
	102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276
	103	Jayshree	Female	17	Jaipur	B.Tech AIML	85.50	9876543276
	104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276
	105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
	106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276
	107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211
	108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
	109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213
	110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214
	111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215
	112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216
	114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250

18. Update the fees of a particular course.

68 • `SELECT * FROM Course;`

Result Grid |   Filter Rows: | Edit:    | Export

	Course_ID	Course_Name	Fees	Duration
▶	1	B.Tech CSE	90000.00	4 Years
	2	BCA	60000.00	3 Years
	3	B.Tech IT	80000.00	4 Years
	4	MCA	75000.00	2 Years
	5	MBA	70000.00	2 Years

19. Update multiple columns of a student simultaneously.

[illegible]

20. Delete a particular student record using Student_ID.

```
72 • SELECT * FROM Student WHERE Student_ID = 140;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

21. Delete all students belonging to a particular city.

```
73 • DELETE FROM Student WHERE Address = 'Delhi';
```

```
74 • SELECT * FROM Student WHERE Address = 140;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

22. Delete employees whose salary is below a specified amount.

```
75 • DELETE FROM Employee WHERE Salary < 30000;
```

23. Delete products whose quantity is zero.

```
76 • DELETE FROM Product WHERE Quantity = 0;
```

24. Delete records satisfying multiple conditions using AND/OR.

```
77 • DELETE FROM Student WHERE Age > 21 AND Percentage < 70;
```

```
78 • DELETE FROM Student WHERE Address = 'Delhi' OR Address = 'Mumbai';
```

25. Perform a complete sequence of **INSERT**, **UPDATE** and **DELETE** operations on a Student table and display the table after each operation.

DISPLAY initial table:

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216

INSERT the student:

```
81 • (150, 'Rahul', 'Male', 21, 'Jaipur', 'BCA', 80.50, '9876543260');
```

```
82 • SELECT * FROM Student;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250
141	Arjun	Male	21	BCA	BCA	70.50	70505
150	Rahul	Male	21	Jaipur	BCA	80.50	9876543260

UPDATE the student:

```
84 WHERE Student_ID = 150;  
85 SELECT * FROM Student;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wr

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250
141	Arjun	NULL	21	NULL	BCA	NULL	NULL
150	Rahul	Male	21	Delhi	BCA	85.00	9876543260
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

udent 34 x

DELETE the student:

```
86 DELETE FROM Student WHERE Student_ID = 150;  
87 SELECT * FROM Student;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Cont

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250
141	Arjun	NULL	21	NULL	BCA	NULL	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Instructions:

1. Implement each task in a separate query.

Submission:

1. Take screenshots of each step's output.
2. Save the SQL scripts in a text file.